

ZACHARY LUKAN

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EDUCATION

McGill University, Montreal

Graduating class of 2029

B.S. in Computer Science, Minor in Math

Related Coursework: Object-Oriented Programming in Java, Computer Systems, Software Systems, Statistics and Probability, Discrete Mathematics, Multivariate Calculus

LEADERSHIP EXPERIENCE

Canadian CanSat Launch Campaign

June 2025

Lead of Software Development

Lethbridge, AB

- Collaborated with team to design and manufacture a compact FPV drone capable of withstanding XX G force and self-stabilize, connected to VR headset, won technical design award
- Developed using C++ a system to report air pressure, position, and force to the ground station with a range of XX km
- Presented data to panel of judges, reached over 1 million users online

University of British Columbia Physics Olympics

March 2025

Team Lead

Vancouver, BC

- Led a team of developers and engineers to win against 80 teams by over XX percent/points
- Directed competition teams of 7-10 members by managing team meetings and overseeing multiple engineering projects

PROJECTS

Facial Tracking and Emotion Recognition Using Neural Networks

OpenCV, PyTorch, FastAPI, Docker,

NiceGUI, Ax

- Engineered a real-time facial emotion recognition system using OpenCV-based face tracking and a PyTorch CNN, transforming live video streams into emotion classifications with XX% accuracy and ZZ fps
- Developed and deployed a production-ready ML application stack with FastAPI, Docker, and NiceGUI, exposing model inference through an interactive web interface accessible from any device
- Increased model performance through Bayesian hyperparameter optimization using Ax, automating experimentation across training configurations and accelerating convergence resulting in a YY% increase in performance

Panorama Stitcher with Blending

JavaScript, CSS, Python, OpenCV, Django

- Integrated a backend using
- Processed and stitched together images using feature matching and Gaussian blending
- Designed a frontend UI with HTML and JavaScript and communicated to the backend using Django

Personal Portfolio Website

JavaScript, CSS, HTML

- Integrated a backend using
- Processed and stitched together images using feature matching and Gaussian blending
- Designed a frontend UI with HTML and JavaScript and communicated to the backend using Django

TECHNICAL STRENGTHS

Computer Languages

Java, Python, C/C++, Assembly, JavaScript

Frameworks

OpenCV, PyTorch, NumPy, Pandas, Django

Technical

Data Structures and Algorithms, Computer Vision, Machine Learning, Neural Networks

Tools

Git, GitHub, Google Cloud, Microsoft Office, Google Colab, Visual Studio Code